

Section-A (25 Marks)

1. Discuss the physical characteristics of the STOL Port, its obstacles and limitations. [5]

Answer:

STOL Port (Short Take-Off and Landing port) is a specialized airfield designed for fixed-wing aircraft that require minimal runway length. STOL ports are incredibly practical for mountainous terrain or highly congested cities where space for a standard, sprawling airport is unavailable.

In Nepal, some of the notable operational STOLport are Tenzing Hilary (Lukla) Airport, Talcha Airport (Mugu), Simikot Airport (Humla) and many others in Himalayan region.



The physical characteristics of the STOL Port are as follows:

- i. **Runway length and width:**
STOL ports use shorter runways than conventional airports, typically between 600 meters and 1000 meters.
- ii. **Surface and pavement:** The surfaces are designed to provide adequate braking performance. Rough textures or grooved surfaces may be used to improve friction and water drainage.
- iii. **Approaches and obstacle clearance:** STOL ports often implement inclined takeoff/landing surfaces to manage obstacle clearance and to minimize airspace needs in terminal areas.

Obstacles of STOL Port:

- i. **Terrain:**
Mountainous STOL ports (like Lukla) are often carved into rugged terrain, requires high-precision flying to navigate around tall trees, peaks, and ravines.
- ii. **Short length constraints:**
Operating within very short strips (often under 800 meters) means there is zero room for error, requiring exceptional short-field capabilities.
- iii. **Wind and turbulence:**
Mountainous environments are highly susceptible to unpredictable crosswinds, wind shear, and downdrafts, creating challenge for STOL port Operations.

Limitations of STOLport:

- i. **Types of aircraft:**
Only those aircraft with strong STOL performance can use STOLports. Short Runway length restrict maximum takeoff weight excluding the large jet aeroplanes to carry out flying operation in STOLports.

- ii. **Regulatory ambiguity:**
The classification, certification standards, and procedures are complex or underdeveloped. International Civil Aviation Organization (ICAO) lacks binding Standards and Recommended Practices (SARPs) for STOL ports.
- iii. **Weather and terrain:**
short runways and steeper approach paths make flying operations more sensitive to wind and low visibility in STOLport.

2. Introduce the Convention on International Civil Aviation, also known as the Chicago Convention. [5]

Answer:

Convention on International Civil Aviation (also known as Chicago Convention), was signed on 7 December 1944 by 52 States. ICAO came into being on 4 April 1947. The Convention on International Civil Aviation set forth the purpose of ICAO to help create and preserve friendship and understanding among the nations and peoples of the world. Nepal obtained the membership of International Civil Aviation Organization (ICAO) in 1960.

Key legal provisions:

- i. **Sovereignty over airspace:** Each state has complete and exclusive sovereignty over the airspace above its territory.
- ii. **Freedoms of air:** The convention underpins the concept of “freedoms of the air”. The first and second freedoms apply automatically to contracting states, while further freedoms depend on bilateral or multilateral agreements.
- iii. **Licensing and Certificates:** It requires states to recognize certain certificates and licenses issued under other contracting states such as pilot licenses, operator certificates.

Standards and Recommended Practices (SARPs)

- i. The Convention empowers ICAO to develop Annexes consisting of Standards and Recommended Practices (SARPs) including the subjects of aircraft operations, airworthiness, air traffic services, aerodromes, and personnel licensing.
- ii. SARPs aim to harmonize national regulations with international practices and standards so international operations can be conducted safely and efficiently.

Operational and regulatory impact

- i. **Air navigation and safety:** The Convention and ICAO SARPs form the baseline for global air navigation rules, air traffic management, and aviation safety oversight.
- ii. **Facilitation and border processes:** It includes provisions to ease international air transport, such as passport, customs, and public health arrangements, standardizing the arrival/departure formalities.
- iii. **Economic and bilateral effects:** While the Convention provides a legal framework, access rights and commercial routes are governed largely by bilateral or multilateral air services agreements built on the Convention’s framework.

3. Unlike in Working Stress Method, discuss how a different kind of approach is adopted while using safety factors to take care of uncertainty in material strength and load in the case of Limit State Design of RC Structure. Explaining under-reinforced and over-reinforced RC sections, clarify how and why an over-reinforced section is not advisable to provide. [6+2+2=10]

Answer:

In the Working Stress Method (WSM), uncertainties in material strength and loads are accounted by using a single overall factor of safety (FOS). The design load is taken equal to actual working loads. It is a deterministic method which assumes that the stresses due to working load don’t exceed the safe permissible stresses.

$$\text{Permissible Stress} = \text{Characteristic Strength} / \text{Factor of Safety}$$

Working stress approach does not distinguish between uncertainties arising from loads and those arising from material strengths.

Meanwhile, Limit State Design (LSD) adopts more rational and realistic approach by considering uncertainties in loads and material strengths separately through partial safety factors.

- i. **Safety factors for loads:**
Characteristic loads are multiplied by load factors to obtain the design load.

Design Load = $\gamma_f \times$ Characteristic Load

Where γ_f = Partial Safety Factor for Load

This accounts for possible increase in loads such as dead load, live load, earthquake loads etc.

ii. Safety factors for materials:

The characteristic strengths of materials are reduced by partial safety factors:

$$f_{cd} = \frac{f_{ck}}{\gamma_m}$$
$$f_{yd} = \frac{f_y}{\gamma_m}$$

where:

f_{ck} = characteristic compressive strength of concrete

f_y = characteristic yield strength of steel

γ_m = partial safety factor for material

Typical values (as per design codes):

Concrete: $\gamma_m=1.5$

Steel: $\gamma_m=1.15$

Advantages of LSD over WSM:

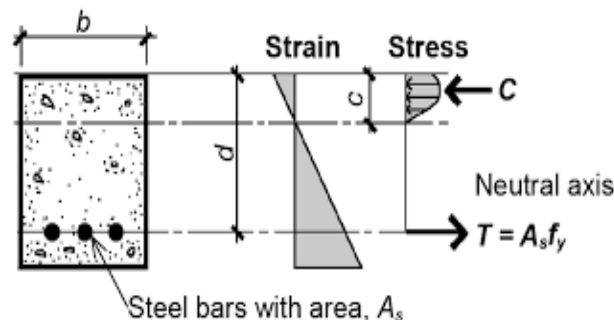
- i. LSD is more realistic and economical design.
 - ii. Ensures both safety and serviceability requirements.
 - iii. Reflects actual ultimate strength behavior of reinforced concrete.
 - iv. Consideration of load and material uncertainties separately.
- Hence, limit state design (LSD) allows uncertainties in loading and material properties to be considered independently, resulting in a safer, more economical, and scientifically sound design of RC structures.

Under-reinforced section:

- The section in which maximum stress in the steel reaches before that of concrete is known as under-reinforced section.
- In this type of section, steel fails before concrete.
- Depth of neutral axis is less than that of balanced section.
- Failure of under reinforced section is ductile or tensile in nature.

iii. Over-reinforced section:

- The section in which maximum stress in the concrete reaches before that of steel is known as over reinforced section.
- In this type of section, concrete fails before steel.
- Depth of neutral axis is higher than that of the balance section.
- Failure is sudden and brittle with little warning.



An over-reinforced section is not advisable to provide because of the following reasons:

- a. Failure in over-reinforced section is brittle in nature where large deflection and cracks occur without giving sufficient warning about the occurred failure.

- b. Over-reinforced section lacks ductility so that it doesn't provide sufficient time for the occupant to vacate the structure and take precautionary measures before complete failure occurs.
- c. The design of over-reinforced section is uneconomical as excess amount of steel is provided which cannot be fully utilized because concrete fails before steel reaches yield stress.

4. What are the factors that govern the size of the Apron-Gate position in an airport? Also, sketch different basic aircraft parking configurations and discuss advantages and disadvantages of each of them? [10]

Answer:

Apron is a designated area paved for parking, loading, unloading, refueling and boarding purposes of the aircraft. Apron gate is space allotted to the aircraft for parking and other operations at the apron. There are several factors that govern the size of apron-gate position in an airport. They are described in brief:

i. Size of aircraft:

The gate must physically accommodate the largest aircraft expected to use it. Larger aircraft need more stand width, longer clearance space, and more room for wingtip separation.

ii. Minimum turning radius required:

The gate size should provide enough space for the aircraft to maneuver safely in and out of the apron.

iii. Safety and operational clearances:

Space must be designated to keep the path clear of ground equipment and fixed structures. Defined distances must be maintained between parked aircraft to avoid collisions.

iv. Ground service equipment and servicing:

The gate size is heavily influenced by the ground operations required during a turnaround. The position of the gate is also calculated to align perfectly with the reach and operational angle.

Hence, the apron gate position is governed by aircraft size, number of gates required, along with safety clearances and ground-handling needs.



Different basic aircraft parking configurations:

Aircraft parking configurations refer to how an aircraft is oriented relative to the terminal building. There are five most basic aircraft parking configurations. They are listed and sketched below:

i. Parallel parking:

This parking configuration is used for remote stands and for large cargo operations. Passengers are usually transported from the terminal to the aircraft using buses or walk.

Advantages:

- a. No push-back is usually required in parallel parking.
- b. Easy maneuvering of aircraft.
- c. Safety during aircraft movement.

Disadvantages:

- a. Requires a very large apron area.

- b. Long walking distance for passengers.

ii. **Nose-in parking:**

Aircraft is parked perpendicular to the terminal building with its nose facing the structure. It is **the most common aircraft parking configuration** for commercial passenger jets.

Advantages:

- a. Efficient use of apron space.
- b. Suitable for large airports with high traffic.
- c. Short walking distance for passengers.

Disadvantages:

- a. In nose in parking, aircraft requires push-back tractor to leave the stand.
- b. Higher operational cost due to push-back equipment.

iii. **Angled nose-out parking:**

Aircraft is positioned at an angle with its nose pointing away from the terminal. The configuration allows the aircraft to power out without needing a tow-truck.

Advantages:

- a. No-push back tractor is required in angle-nose out parking configuration.
- b. Better visibility to pilot during taxi-out.
- c. Reduced apron congestion due to quicker gate clearance.

Disadvantages:

- a. Angled nose-out parking requires larger apron area than nose-in parking.
- b. More complex apron layout and gate design.

iv. **Angled nose-in parking:**

The aircraft is parked at an acute angle with its nose facing towards the terminal building. This parking configuration allows the aircraft to maneuver into the stand under its own power.

Advantages:

- a. Angles nose-in parking has easier entry and exit than perpendicular nose-in parking.
- b. Reduced push-back distance.
- c. Better visibility for pilots during maneuvering.

Disadvantages:

- a. Requires more apron area than perpendicular parking.
- b. Complex apron layout.

v. **Nose out parking:**

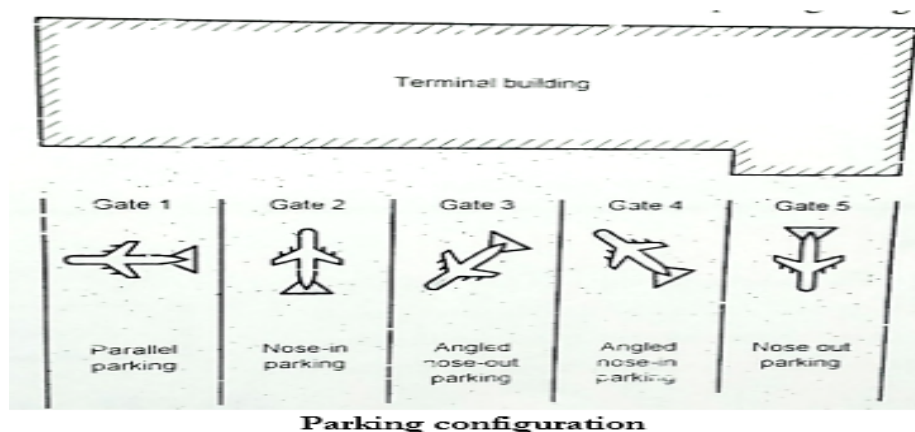
In nose-out parking, airplanes must either be backed into the parking position using a tow tractor or perform a complex turn in front of the gate to align themselves.

Advantages:

- a. No need for push-back equipment in nose out parking operations.
- b. Reduced risk of delays caused by push-back operations.
- c. Simplifies the aircraft departure procedures.

Disadvantages:

- a. Baggage handling and service operations are less convenient.
- b. Less commonly used at major passengers' terminals.
- c. Jet blast from departing aircraft creates safety hazards.



Section-B (25 Marks)

5. Elaborate the different elements of airport planning. Explain the 'centralized' and 'decentralized' concept of terminal building planning with their salient features. [6+4=10]

Answer:

Airport planning is the systematic process of determining the requirements, layout, facilities, and future development of an airport to ensure safe, efficient, and economical air transportation.

The major elements of airport planning are:

- i. **Air traffic demand forecasting:**
 - Estimation of future passenger, cargo, and aircraft movements is done.
 - Analysis of population growth, economic development, tourism, and regional transportation needs is also carried out in air traffic demand forecasting.
 - This forecasting forms the basis for determining airport capacity and expansion requirements.
- ii. **Site selection:** Different factors are analyzed for site selection.
 - Topography and soil conditions.
 - Meteorological conditions (wind, visibility, fog)
 - Accessibility to urban centers.
 - Availability of land for future expansion
- iii. **Planning of airport facilities:** It includes facilities directly related to aircraft operations:
 - Number, orientation, length, width, and pavement design of runways is planned.
 - Layout of taxiways is also appropriately planned for safe and efficient aircraft movement.
 - Aprons facilities for parking, loading, fueling, and servicing areas are provided for preventing delays.
 - Planning of lighting systems, instrument landing systems, radar, etc. for navigational purposes should also be carried out.
- iv. **Planning of terminal facilities:**
 - Planning of passenger terminal buildings must be carried out.
 - Provision of check-in counters, baggage handling systems should be provided.
 - Customs, immigration and waiting lounges are also appropriately planned.
- v. **Planning of airport area:** it includes the facilities connecting airport with surrounding areas such as:
 - Roads and highways access to the airport.
 - Parking facilities for vehicles.
 - Public transportation facilities and taxi services.
- vi. **Airport capacity planning:**
 - Planning of runway, taxiway, apron, and terminal capacities is done for this step of airport planning.
 - Assessment of peak-hour traffic demand must be done to estimate and design the facilities in airport.
 - Measures to reduce congestion and delays must be properly studied.
- vii. **Environmental considerations:** in airport planning, environmental considerations that must be analyzed are:
 - Noise impact assessment.
 - Air and water pollution control.
 - Waste management systems.
 - Environmental Impact Assessment (EIA) and sustainability measures.
- viii. **Safety and security planning:** It includes:
 - Aircraft operational safety.
 - Fire-fighting and rescue facilities.
 - Security screening systems.

- Emergency response and disaster management plans.
- ix. **Utility and support services planning:** The planning includes:
 - Water supply and drainage systems.
 - Power supply and backup systems.
 - Communication and navigation facilities.
 - Fuel storage and distribution systems.
 - x. **Financial and economic planning:**
 - Estimation of construction, operation, and maintenance costs is determined to carry out effective airport planning.
 - Revenue forecasting from passengers, cargo, concessions, and parking can also be carried out to mark the profit or loss made.
 - Cost-benefit and feasibility analyses must be carried out for efficient planning.
 - xi. **Master planning and future expansion:**
 - Preparation of an airport master plan for 20–30 years.
 - Identification of future runway, terminal, and apron expansion areas.
 - Phased development to accommodate future growth.

Terminal building planning is a process of making strategy based on how passenger-processing facilities such as check-in, security, baggage handling, customs, and boarding gates are arranged within the airport. Centralized and decentralized terminal building concepts are the two terminal building philosophies in aviation infrastructure. Centralization consolidates passenger processing (ticketing, security, baggage) into a single main hub. Decentralization divides these functions into smaller, independent units situated directly adjacent to specific aircraft gates.

Centralized terminal building concept:

- i. In this concept, Passengers complete all formalities in one central location before proceeding to their aircraft gates.
- ii. Salient features:
 - Entire passengers and airline are served by single terminal building.
 - Check-in, baggage handling, security, customs and immigration are centralized.
 - Transfer of passengers between airlines is convenient.

Decentralized terminal building concept:

- i. In this concept, the airport functions are distributed among separate terminal buildings or terminal units, each serving specific airlines, groups of airlines, or operational areas.
- ii. Salient features:
 - It consists of multiple terminal buildings.
 - Each terminal is provided with passenger processing facilities.
 - This concept encourages reduction of congestion in single building.
 - Suitable for large international airports.

6. For the purpose of identifying standards for various sizes of airports and the functions they serve, reference codes have been developed by ICAO. Elaborate Aerodrome Reference Code as per ICAO Annex 14. [4+6=10]

Answer:

The Aerodrome Reference Code is a coding system developed by the **International Civil Aviation Organization (ICAO)** under Annex 14 – Aerodromes which provides a standardized method that relates airport physical characteristics to the aircraft intended to use the aerodrome.

The code consists of two elements:

- i. Code Number (1, 2, 3, or 4):
It is based on the aircraft reference field length.
- ii. Code Letter (A to F):
It is based on aircraft wingspan and outer main gear wheel span.

The Aerodrome Reference Code is written as:

Code Number + Code Letter

For example: **2B, 3C, 4E, 4F**

i. Code Number:

Code number represents the minimum length of runway required for aircraft to take off at sea level under standard atmospheric conditions. These code numbers are determined by Aeroplane Reference Field Length (ARFL).

Code number according to Aeroplane Reference Field length (ARFL):

Code Number	Aeroplane Reference Field Length (ARFL)
1	< 800 m
2	800-1200 m
3	1200-1800 m
4	1800 or more

ii. Code Letter:

From the wingspan of aircraft and outer main gear wheel span (OMGWS), the code letter is determined.

Code Letter	Wingspan (m)	Out main gear wheel span (m)
A	Up to but not including 15	Up to but not including 4.5
B	15 to < 24	4.5 to < 6
C	24 to < 36	6 to < 9
D	36 to < 52	9 to < 14
E	52 to < 65	9 to < 14
F	65 to < 80	14 to < 16

Some of the typical aircraft and corresponding codes:

- a. Boeing 737 = 4C
- b. Airbus A320 = 4C
- c. Boeing 767 = 4D
- d. Boeing 777 = 4E
- e. Airbus A380 = 4F

Aerodrome Reference Code is used to determine:

- i. Runway width
- ii. Taxiway width
- iii. Shoulder requirements
- iv. Apron dimensions
- v. Airport geometric design standards

Thus, the Aerodrome Reference Code of ICAO Annex 14 is a two-part designation that ensures all airport facilities are compatible with the size and performance characteristics of the aircraft using the aerodrome. It helps to establish relevant design standards for runways, taxiways, aprons and other airport facilities. Hence, the code forms the basis of planning and design of airport facilities worldwide.

7. What is understood by Float in Critical Path Method (CPM). Explain with examples. [5]

Answer:

The amount of time by which an activity can be delayed without affecting the project completion time or schedule of subsequent activities is known as float. It is also called slack. It denotes to the availability of flexibility in the schedule of project activities.

So, in critical path, activities have zero float and noncritical activities has positive float.

There are four types of floats. They are:

i. Total float:

It is the maximum time by which an activity can be delayed without delaying the overall project completion time.

$$TF = LS - ES = LF - EF \quad \text{Where:}$$

ES = Earliest Start

EF = Earliest Finish

LS = Latest Start
LF = Latest Finish

Example:

Suppose an activity has ES= 5 days, EF= 10 days, LS= 8 days and LF= 13 days. Then,
Total float = LS- ES = 8-5= 3 DAYS
= LF-EF = 13-10 = 3 DAYS

ii. Free float:

It is the amount of time an activity can be delayed without delaying the earliest start of any succeeding activity.

$$FF = ES_{\text{SUCCESSOR}} - EF_{\text{ACTIVITY}}$$

Example:

Let us consider that an activity A completed in 10 days. If its successor activity starts at 12th day, then

$$\begin{aligned}\text{Free float} &= ES_{\text{SUCCESSOR}} - EF_{\text{ACTIVITY}} \\ &= 12-10 = 2 \text{ DAYS}\end{aligned}$$

Hence, we can say that in this example, activity A can be delayed by 2 days without affecting B.

iii. Independent float:

it is defined as the amount of time by which an activity can be delayed without affecting succeeding or preceding activities.

$$IF = ES_{\text{SUCCESSOR}} - LF_{\text{PREDECESSOR}} - \text{DURATION}$$

If the value is negative, Independent Float is taken as zero.

Example: let us consider that LF_{PREDECESSOR} = 8 days, duration of activity= 5 days and ES_{SUCCESSOR} =15 days. Then,

$$\begin{aligned}\text{Independent Float} &= IF = ES_{\text{SUCCESSOR}} - LF_{\text{PREDECESSOR}} - \text{DURATION} \\ &= 15 - 8 - 5 = 2 \text{ days}\end{aligned}$$

iv. Interfering float:

The difference between total float and free float is known as interfering float.

$$\text{Interfering Float} = TF - FF$$

Example: From above example, interfering float= 3-2 = 1 DAY

Hence, float in CPM helps to identify critical and non-critical activity and assists in project scheduling and allocation of resources.

8. Explain the basic tests for coarse aggregate material to ensure its suitability to use in base course of airport pavement. [5]

Answer:

The base course of an airport pavement must withstand heavy wheel loads, repeated aircraft movements, abrasion, and weathering effects. Therefore, coarse aggregates used in the base course should possess adequate strength, toughness, durability, and proper grading.

The following tests are commonly conducted to assess the suitability of coarse aggregates:

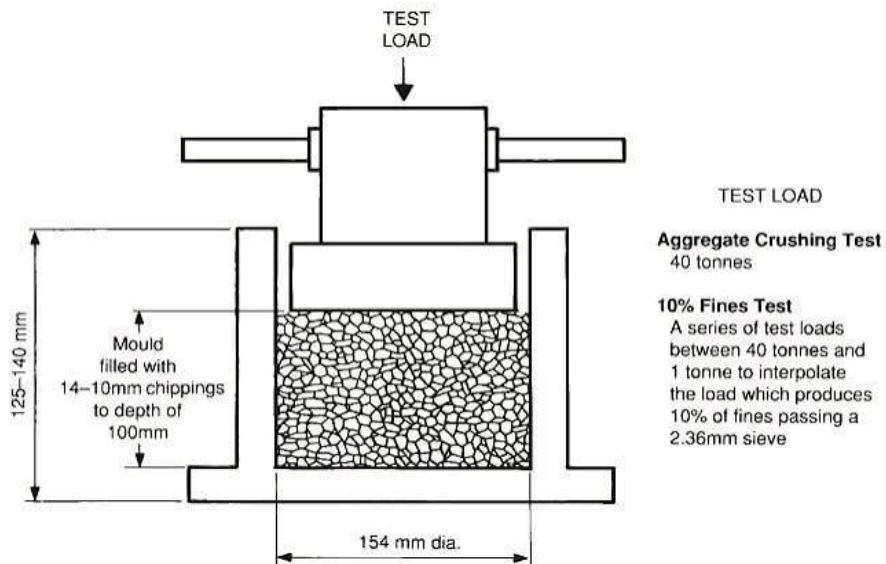
i. Aggregate crushing value test (ACV Test):

This test determines the resistance of aggregates to crushing under gradually applied compressive loads which indicates the strength of aggregates.

ACV value is given by the formula:

$$ACV (\%) = \frac{\text{weight of fines}}{\text{total weight of samples}} \times 100\%$$

Lower value of ACV indicates stronger aggregate. Hence, for airport pavements, aggregate with low crushing values are preferred.



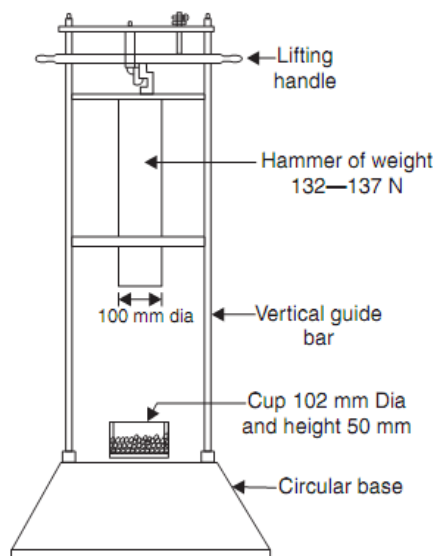
ii. Aggregate impact value test:

Aggregate impact value gives the measure of toughness of the aggregates against sudden impact or shock loads.

AIV value is given by the formula:

$$\text{AIV (\%)} = \frac{\text{weight of fines}}{\text{original weight of sample}} \times 100\%$$

Lower value of AIV indicates tougher aggregate. Hence, for airport pavements, aggregate with lower AIV is preferred.



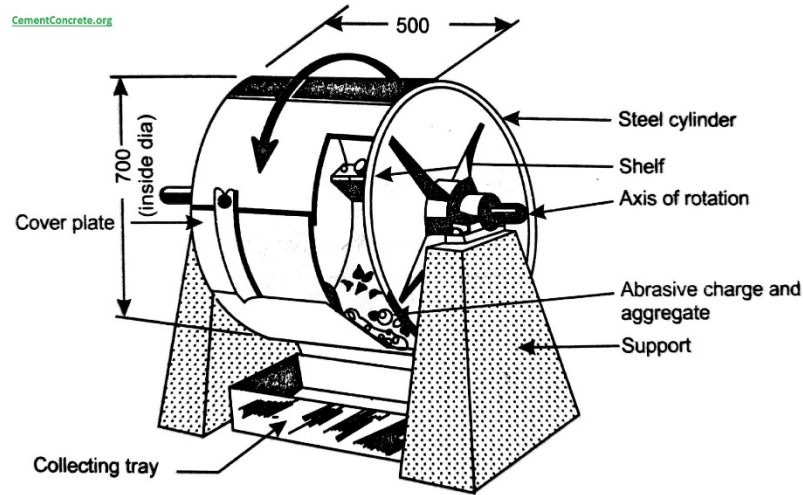
iii. Los Angeles abrasion test:

This test determines the resistance to abrasion, wear, and attrition.

LA value is given by the formula:

$$\text{LA(\%)} = \frac{\text{original weight} - \text{final weight}}{\text{original weight of sample}} \times 100\%$$

Lower abrasion value indicates higher resistance to wear. Hence, the aggregate with lower abrasion value is used for construction of airport pavements.



iv. Soundness test:

Soundness test evaluates the resistance of aggregate to weathering actions such as alternate wetting, drying, freezing, and thawing. The test ensures durability of aggregate under environmental and climatic conditions.



v. Specific gravity and water absorption test:

This test helps to determine porosity and density of aggregates.

The formula to calculate specific gravity and water absorption of aggregates are as follows:

$$\text{Specific Gravity} = \frac{\text{weight of aggregate in air}}{\text{weight of equal volume of water}}$$

$$\text{Water absorption} = \frac{W_2 - W_1}{W_1} \times 100 \%$$

vi. Gradation test:

Also known as sieve analysis which determines particle size distribution and check the grading of aggregate, whether it is well graded or not for a strong compactable base layer.



9. Differentiate between soil compaction and consolidation. Example how each of these processes affects the structure. [5+5=10]

Answer:

Soil compaction is the process of densification of soil by removing the air voids present in the soil particles under the action of mechanical energy. Similarly, consolidation is the process of compression of water voids due to which expulsion of water from the void occurs under the effect of applied load.

The differences between soil compaction and consolidation are given in points.

	Compaction		Consolidation
1.	Densification of soil under the action of mechanical energy.	1.	Compression of water voids under the effect of sustained applied load.
2.	Compaction is immediate process.	2.	Consolidation is slow and time-dependent process.
3.	It occurs due to reduction of air voids.	3.	It occurs due to reduction of water filled voids.
4.	Compaction increases dry density and strength of soil.	4.	Consolidation causes settlement and increase in effective stress.
5.	Compaction is generally artificial and intentional process.	5.	Consolidation is generally natural or load-induced.
6.	Equipment such as roller, rammer, vibrators are used for compaction.	6.	No mechanical equipment is required for consolidation. It occurs under structural load.



Compaction at optimum water content (OMC) is done before construction of any structures such as highway pavement or airport runway or foundations of buildings, by using mechanical equipment such as rollers and rammer. The process of compaction increases soil density and bearing capacity. It also reduces the settlement of the structure or pavement that might occur in the future due to long term settlement. The pavement with improper compaction suffers from uneven settlement, formation of ruts and corrugations, cracks and premature failure of pavement under traffic loads. The foundation with improper settlement makes the foundation weak and prone to failure when subjected to dynamic loads such as earthquake.

For example: An airport runway is built on poorly compacted soil. After repeated aircraft maneuvers and other operations, some parts of runway show the sign of damage, causing uneven settlement, surface undulations. This increases the cost of repair and maintenance works and might cause accidents or mishaps due to the poor quality of runway construction. But on the other hand, if proper compaction is to be done before hand while construction, then it improves the performance and stability of structure by preventing excessive settlement.

Consolidation is the long-term process in which removal of water from the voids occurs under the sustained applied load. The primary effect of consolidation is occurrence of ground settlement. If the settlement occurs uniformly, the structure will settle slightly without major structural damage. However, differential settlement

across a foundation causes different parts of foundation to sink at different rates, leading to severe twisting, structural cracking and foundational instability.

For example: A multistoried building is constructed on thick layer of saturated clay. As the building load increases, stress on the clay also increases. Consolidation phenomenon slightly arises with water slowly escaping from the soil pores. The clay gradually compresses over months and years, causing ground surface to settle gradually and slowly. Equal or unequal settlement might occur. Equal settlement may create disconnection with utilities, cracks on the non-structural component of structure. However, differential settlement leads to development of cracks may develop in walls, floors and beams causing structural damage in the structure.

10. Explain the design considerations for airport drainage systems, including the determination of the design storm and layout of surface and subsurface drainage. [6+4=10]

Answer:

Airport drainage systems are the systems designed to remove surface water and groundwater immediately from runways, taxiways, aprons, and surrounding airport areas. Proper drainage is an essential requirement in airport for maintaining pavement strength, ensuring aircraft safety, and preventing waterlogging and also increasing service life of pavements and runways. It also helps to reduce the frequency of accidents in the airport periphery due to slippery runways and reduce the quantity of repair and maintenance works.



Design considerations for airport drainage systems:

- i. Selection of adequate design storm:**
 - Rainfall intensity using local IDF (Intensity-Duration-Frequency) data is determined.
 - An appropriate return period (typically 5–25 years depending on facility importance) is selected.
 - The drainage system to safely carry peak runoff is designed.
- ii. Rapid removal of surface water:**
 - Ponding on runways, taxiways, and aprons must be prevented by drainage systems.
 - Sufficient transverse and longitudinal slopes must be provided.
 - Quick runoff to side drains and storm sewers must be ensured.
- iii. Protection of pavement structure:**
 - Airport drainage systems must prevent infiltration of water into pavement layers.
 - The strength of the subgrade, base, and subbase must be maintained.
 - The pavement deterioration and maintenance costs must be reduced.
- iv. Efficient surface drainage layout:**
 - Proper provision of side drains, catch basins, culverts, and stormwater channels.
 - Smooth flow paths without obstruction must be ensured.
 - Water accumulation near pavement edges must be avoided.
- v. Provision of subsurface drainage:**
 - Underdrains and perforated pipes must be installed where groundwater levels are high.
 - Seepage water from beneath pavements must be removed.

- Softening of the subgrade must be prevented.
- vi. **Soil and groundwater conditions:**
- Permeability of soil and groundwater level must be considered.
 - Suitable filter materials around subsurface drains should be provided.

Determination of design storm:

i. **Return period:**

Return period for different drainage facilities are:

Facility	Return period
Surface drains	5-10 years
Main drainage channel	10-25 years
Critical airport facility	25-50 years

ii. **Intensity of rainfall:**

Rainfall intensity is determined from IDF curve or rational and empirical formulas.

iii. **Catchment area:**

The area contributing runoff to the drainage system is known as catchment area. The design of surface and subsurface drainage system largely depends upon the extent of catchment area.

iv. **Runoff coefficient:**

It depends upon the type of surface. Highly permeable surface has lesser runoff coefficient and vice versa.

Surface type	Runoff coefficient
Concrete pavement	0.85-0.95
Asphalt pavement	0.8-0.9
Grass areas	0.1-0.4

v. **Runoff calculation:**

Runoff calculation is generally done by using rational method.

$Q = CIA$ where C= coefficient of runoff

I= Intensity of rainfall

A=catchment area

The value of surface runoff helps to design the drainage channels.

Layout of surface drainage:

i. **Pavement cross slope:**

Transverse slope must be provided in airport runways, taxiways and apron which allows rapid draining of rainfall runoff.

Typical slope of:

- Runway =1-1.5%
- Taxiway =1-1.5%
- Apron = 1-2%

ii. **Longitudinal slope:**

Longitudinal slope allows continuous flow of runoff towards the collection drains which prevents water to stand on the runways, taxiways and apron.

iii. **Side drains:**

Open channels along the runway and taxiway edges are provided which is known as side drains. This collects the surface runoff immediately from the pavement surfaces.

iv. **Catch basins and inlets:**

These are connected to underground storm sewers. They receive surface runoff from the paved areas.

Typical surface drainage arrangement:

Runway Surface
↓
Cross Slope
↓

Side Drain
↓
Storm Sewer
↓
Outfall Channel

Layout of subsurface drainage:

- i. **Perforated drain pipes:**
These drain pipes are installed underground below the pavement edges and collect the groundwater whose level is near the pavement subgrade.
- ii. **Filter layers:**
They are the layer of granular material surrounding the drain pipe to prevent soil particles from entering the pipe.
- iii. **Longitudinal underdrains:**
Longitudinal underdrains run along the runway and taxiways.
- iv. **Transverse and collector drains:**
Transverse drains carry the runoff to collector drains and collector drains carry the water to disposal points.

Typical subsurface drainage system:

Pavement

Base Course

Subbase

Perforated Pipe

Filter Material

Natural Soil

11. What is pavement management system? Explain different types of distress in bituminous pavement of an airport. [3+7=10]

Answer:

A pavement management system is a structured way to monitor, evaluate, and plan maintenance and rehabilitation for roads, streets, runways, and other paved assets. It helps owners decide where to spend limited funds to keep pavements in good condition for as long as possible.

Importance of pavement management system:

- i. It helps to collect the data of pavement conditions through inspection.
- ii. It uses the data to rate the pavement health by using indexes such as PCI (Pavement condition Index)
PCI = 85–100: Excellent
PCI = 70–85: Good
PCI = 55–70: Fair
PCI = Below 55: Poor to Failed
- iii. It predicts the deterioration rate and need for repair and maintenance of pavements.
- iv. The system helps to prioritize repairs, preservation, resurfacing and reconstruction.

Different types of distress in bituminous pavement of an airport:

- i. **Alligator cracking:**
These cracks are caused by repeated aircraft wheel loads and structural failure of the pavement. Generally, they occur along the wheel paths. These cracks lead to rapid deterioration of pavement.



ii. Longitudinal cracking:

Longitudinal cracks develop parallel to the u=runway or taxiway centerline. These types of cracking are caused by poor construction joints, differential settlement, shrinkage and others.



iii. Rutting and corrugations:

Ruts are the permanent longitudinal depressions formed along the wheel paths. It is caused due to excessive aircraft loads, weak subgrade, poor compaction or insufficient pavement thickness. Corrugations are the series of transverse ridges and depressions caused due to unstable asphalt mix and aircraft braking and acceleration forces.



iv. Shoving:

It is the localized displacement of pavement materials caused due to heavy braking of aircraft and poor mix stability.

v. Depression and settlement:

It is the localized low areas on the pavement surface caused due to poor subgrade, poor compaction and consolidation of underlying soil.

vi. Raveling:

it is a progressive loss of aggregate particles from the pavement surface, raveling is caused due to insufficient bitumen, aging and oxidation of binder and poor aggregate-binder adhesion.

vii. Stripping:

It is the failure caused by separation of bitumen from the aggregate. It is caused due to poor aggregate-bitumen bonding.

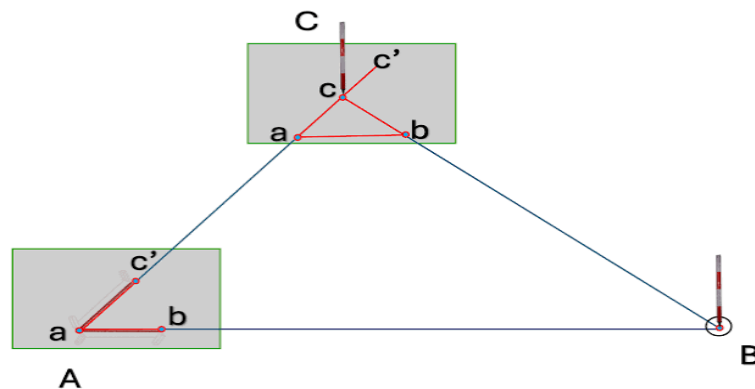
viii. Edge cracking:

Cracks occurring near the pavement edge is known as edge cracking. It is caused due to lack of lateral support, settlement of shoulder and heavy loads near pavement edge.

12. What is method of resection in the context of plane table surveying? Can you correlate resection with working of GPS? Also, explain Reciprocal Levelling and different orders of Benchmarks. [3+2+2+3=10]

Answer:

Resection in plane table surveying is a method of orientation where a surveyor sets up a plane table at an unknown location and determines its exact position on the drawing sheet by sighting multiple known, previously plotted points.



There are three methods of resection. They are:

Methods of resection:

i. Resection by Back Ray:

This method of resection is used when only one known point is available. Here, the plane table is oriented by back sighting. The position is obtained by drawing a ray from the known plotted point.

ii. Two-point problem:

In two-point problem, two known points are visible and already plotted. In this method, an auxiliary station may be required. However, the method is less accurate than three-point method.

iii. Three-point problem:

This method of resection is the most common and accurate method. In three-point problem, three known ground points (A, B, C) and their plotted positions (a, b, c) are used. The table is so oriented that the rays drawn from a, b, c pass through A, B and C respectively. The intersection gives the station position.

Correlation of resection with working of GPS:

Similarities:

Both methods determine an unknown position from known reference points.

- In resection, the surveyor's position is found using known stations A, B, and C.
- In GPS, the receiver's position is found using satellites whose positions are precisely known.

Differences:

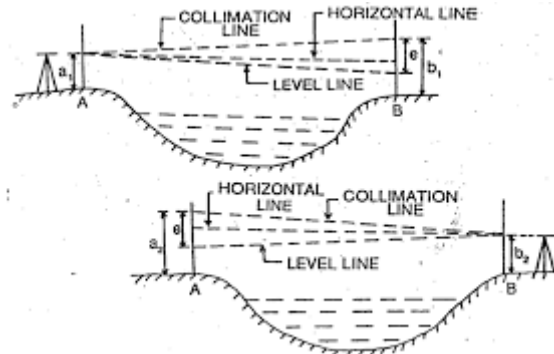
- Resection uses angular observations and graphical construction on a plane table.
- GPS uses distances calculated from radio signals and satellite timing data through trilateration.

Resection is a plane table surveying method used to determine the position of a survey station by sighting known points whose locations are already plotted on the map. It is conceptually similar to GPS positioning, as both

determine an unknown position from known reference points; however, resection uses graphical angular observations, whereas GPS uses satellite signals and trilateration.

Reciprocal levelling:

Reciprocal levelling is a method of levelling used to find the difference in elevation between two distant points when direct levelling is difficult, such as across a river or gorge. It eliminates the errors caused by curvature of the earth, refraction, and slight instrument misalignment.



Let A and B be the two points on opposite side of river.

First setup:

Set up the level near station A. Then, staff readings are taken.

On A = a_1

On B = b_1

Second setup:

Shift the level near B. then, take the staff readings:

On A = a_2

On B = b_2

Formula:

The true difference in level between A and B is:

$$h = \frac{(a_1 - b_1) + (a_2 - b_2)}{2}$$

where,

h = Difference in elevation between A and B

a_1, a_2 = Staff readings on A

b_1, b_2 = Staff readings on B

Advantages:

- i. Reciprocal levelling eliminates collimation, curvature and refraction errors.
- ii. It is suitable for levelling across rivers, lakes, valleys.
- iii. Provides highly accurate results.

Different orders of Benchmark:

i. Zero order Benchmark:

It is the highest precision benchmark used for national geodetic control. This is used in very accurate surveying work.

ii. First order benchmark:

It is very precise which is used to establish primary vertical control across large areas.

iii. Second order benchmark:

It is slightly less precise than first order benchmark and is used for regional and secondary control.

iv. Third order benchmark:

It is used for local surveys and ordinary engineering work where moderate accuracy is enough.